CSE2225 Data Structures **Fall 2023** **Student Name:_____**
Quiz 1 **Student ID:_____**

Q2. (25 pts) Consider the following piece of code below:

```
...
count=0;
for(i=0;i<n;i++)
     for(j=1;j<n;j*=3)
          count++;

printf("count=%d",count);
...
```

What value of count is printed, in terms of **n**, at the end of the execution of the piece of code in Fig. Q2?

Figure Q2. Piece of code for Q2

Answer:

n	1	2	3	4	5	...	n
i	0	0,1	0,1,2	0,1,2,3	0,1,2,3,4		0,1,2,....,n-2,n-1
j	-	1	1	1,3	1,3		1, 3, 9,..., 3^k for k=$\lceil \log_3(n) \rceil$
count	0	2	3	8	10		n×$\lceil \log_3(n) \rceil$

$$count = n \times \lceil \log_3 n \rceil$$

Q3. (25 pts) Solve the following recurrence: f(n)= 7*f(n-1) -10 + 3^n*n with the initial condition f(0)=0.

Answer:

f(n) – 7*f(n-1) = 3^n*n – 10
CE: (x – 1)*$(x-3)^2$*(x-1) = $(x-1)^2$*$(x-3)^2$ = 0
Roots: $x_{1,2}$=1, $x_{3,4}$=3
f(n) = c_1*1^n + c_2*n*1^n + c_3*3^n + c_4*n*3^n
calculate f(1), f(2) and f(3) and solve linear equation system to find c_1, c_2 ...
c_1 = -8,25
c_2 = -8,5
c_3 = 8,25
c_4 = -5
f(n) = -8,25 – 8,5 * n + 8,25 * 3^n – 5 * n * 3^n